

# Spectral methods for singular boundaries: a survey

Comparing GYRE (stellar pulsation) and Dedalus v3 (general spectral) against the Liouville SL-anelastic design

Kiriko, Tsinghua University

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## 0. Update (2026-05-03) — Phase 0 ext+ □□□□

□□□□ 2026-05-02 □□□□□ Path A / B / C □□□□, □□ §6.3 □□□□□□□□□□□□. □□ Phase 0 ext+ □□□□□ □□, □□□□□:

**U1. Path A** □□□□□(§5.1, §6.2)□□□□ — **Path A** □□□  $\sigma$  □□.  $\sigma = 3/2$  □□□□□□  $\pi = r^\alpha u$  □□□□□□□□□□ □□, □□□  $N^{-2}$  □□□□□□□□□□□□. SymPy □□□  $\alpha_2^* = 1 - \sigma/2$  □ Liouville-Schrödinger □□□□□  $t^{-2}$  □□, □□ raw self-adjoint □□□□□□ — □□□□□□□□□□ □□ ( $r$ ) □□, □  $(1-r)^\sigma$  □□□□□ Chebyshev □□□□□□□□  $N^{-\sigma-1/2}$  □□(Trefethen Thm 7.2).

**U2. □□ vs □□  $\sigma$  □□:** □□□□□□□ **non-trivial** □□.  $\sigma \in \mathbb{Z}$ (□  $n = 3$  Eddington)□  $(1-r)^\sigma$  □□□□□, raw Chebyshev □□□□□□□□□□□□. □□ Path A □  $n = 3$  □□ □□□□ “raw Chebyshev, □□□□□□□”(□□□□□), □□  $n = 3/2$  □□ □□□ Path B. □□ docs/polytropic\_index\_spectral\_convergence\_2026-05-03.md.

**U3. □□□□□□□ Path A** □□□□: - y □□: **raw Chebyshev collocation**(□ SL □□, □ Jacobi) - □□: Eddington  $n = 3$  polytrope - Path B (Jacobi) □□□□□□□□  $n = 3/2$  □□□□□□□□□□□□ - Path C (□□□□□□□) □□, □□ angle □□ “1D □□ benchmark” □□ “2D anelastic DNS + □□□□□□□”(□ U4).

**U4. Paper angle** □□(§6.1 □□□□□). □ “Unified spectral basis for Poisson + g-mode spectroscopy” angle □ Poisson □ g-mode □□□□□□□(□ docs/reduced\_pressure\_liouville.md □ docs/anelastic\_SL\_spectral\_design\_2026-05-03.nc1)□□□□. □□ angle: “GPU anelastic pseudo-spectral DNS with live eigenmode projection for convection-pulsation coupling” (□ docs/spectral\_stratified\_poisson\_report\_2026-05-03.md §8).

**U5. §6.3** □□□□□□□□□ — **Dedalus Lane-Emden** □□□□□□, □□□□□. E6/E7 □□ raw Chebyshev □ Lane-Emden  $n = 3$  □□  $1.5 \times 10^{-6}$  (□ GYRE full-gravity 4-var), □□□□□□□, Dedalus Jacobi □□ □□□□ benchmark □□□□□.

□□□□□□□□□□□□. GYRE / Dedalus □□□□□(§2, §3)□□□□ □□□□□□□; §5 Path □□□□ §6 □□□□□□□□□□ U1-U5 □□.

# 1. Background and problem statement

## 1.1 Motivation

This project’s `pseudo_spectral` solver currently handles uniform-density 2D incompressible Navier–Stokes on a doubly periodic domain. Extending it to **anelastic stellar convection** promotes the pressure equation from a constant-coefficient Laplace operator to a variable-coefficient elliptic equation:

$$\nabla \cdot \left( \frac{1}{\rho_0(y)} \nabla p \right) = f(x, y)$$

The original design (`docs/anelastic_SL_spectral_design.md`, 2026-05-01) proposes a Liouville normal-form substitution  $\hat{p} = \sqrt{\rho_0} q$  to reduce the variable-coefficient equation to a Schrodinger-type form:

$$q'' + W(y) q - k_x^2 q = g, \quad W(y) = \frac{\rho_0''}{2\rho_0} - \frac{3(\rho_0')^2}{4\rho_0^2}$$

Solving the eigenproblem for  $[d^2/dy^2 + W]$  yields a Sturm–Liouville (SL) basis  $(\mu_n, \psi_n)$ ; all  $k_x$  modes share the same basis, and Poisson inversion reduces to a per-mode scalar division.

## 1.2 The unresolved problem: the $\rho \rightarrow 0$ singularity

Phase 0 validation (`docs/anelastic_sl_phase0_2026-05-02.md`) shows:

- The Lane–Emden  $n = 1.5$  surface  $\rho \rightarrow 0$  drives  $|W| \rightarrow 1.35 \times 10^6$  (divergent).
- Truncation to  $\rho > 0.01$  (i.e.  $r < 0.94$ ) makes the eigenproblem solvable.
- SL–Poisson end-to-end on a manufactured solution:  $\text{err}_{L^2} = 3.7 \times 10^{-6}$ .
- **Convergence order**  $\sim -2.4$  (**algebraic**), not the expected exponential.
- Phase 0 E3 confirms: a smooth Gaussian  $\rho(y)$  restores exponential convergence. The algebraic rate is therefore entirely driven by the singular boundary.

**Core observation.** The Liouville substitution does not eliminate the  $\rho \rightarrow 0$  singularity; it merely transports it from a variable-coefficient elliptic equation to a Schrodinger potential. We still need to address it.

## 1.3 Purpose of this survey

The quantum-mechanics community (Coulomb potentials) and the stellar-pulsation community (adiabatic oscillations) have accumulated 60 years of standard techniques for exactly this class of singularity. This report surveys two representative open-source codes:

### 1. **GYRE** (Townsend 2013; Fortran)

- Astrophysical stellar pulsation (adiabatic oscillations).
- Structurally isomorphic to our problem: a radial SL eigenproblem inside a star.
- Repo: <https://github.com/rhdtownsend/gyre>

## 2. Dedalus v3 (Burns et al. 2020; Python)

- General-purpose spectral PDE framework.
- Covers Chebyshev / Jacobi / Ball / Disk bases.
- Repo: <https://github.com/DedalusProject/dedalus>

The central question: **how do these two communities treat  $\rho \rightarrow 0$ -type boundary singularities, and what can this project borrow?**

## 2. GYRE's strategy

### 2.1 Overall philosophy: pre-designed variable transformations

GYRE does not use a Liouville  $\sqrt{\rho}$  substitution. It employs a **non-dimensional geometric substitution**:

$$y_1 = x^{2-\ell} \cdot \frac{\xi_r}{r}, \quad y_2 = x^{2-\ell} \cdot \frac{P'}{\rho g r}$$

where  $\xi_r$  is the radial displacement and  $P'$  is the pressure perturbation.

**Key point.** This substitution absorbs geometric singularities (origin  $r = 0$  and surface  $\rho \rightarrow 0$ ) **into the definition of the evolved variables**, so that  $y_1, y_2$  remain bounded throughout the radial domain.

*Reference:* docs/source/ref-guide/osc-equations/dimless-form.rst:24-32

### 2.2 Surface boundary: analytic asymptotic form

GYRE does not impose Dirichlet or Neumann conditions at the surface. It uses **physically motivated asymptotic boundary conditions**:

- **DZIEM mode** (Dziembowski 1971):

$$\left\{ 1 + \frac{1}{V}[\dots] \right\} y_1 - y_2 + \dots = 0$$

- **UNNO mode** (Unno et al. 1989): Eddington approximation.
- **VACUUM mode**: minimal form,  $y_2 = 0$ .
- **JCD mode**: Jorgensen-Christensen-Dalsgaard detailed form.

*References:* src/eqns/rad/rad\_obound\_m.fypp:102-133, src/eqns/rad/gyre/OB\_dziem.inc:1-3

These conditions **all assume a physical limit exists at the surface**, rather than a divergence. This is the deep reason GYRE can sidestep the  $\rho \rightarrow 0$  singularity: its non-dimensional variables  $y_1, y_2$  are finite at the surface regardless of what  $W$  is doing.

## 2.3 Grid refinement: ODE-eigenvalue criteria

src/grid/grid\_refine\_m.fypp implements adaptive log-grid bisection. The refinement criterion is driven by ODE eigenvalues  $\chi$ :

```
! Mechanical / thermal / structural criteria combined
split(j) = split_mech(...) .OR. split_therm(...) .OR. split_struct(...)
```

In  $\ln(x)$  space,  $\text{dln}x * |\chi_{\text{imag}}|$  (wavelength criterion) and  $|\chi_{\text{real}}|^{-1}$  (decay length criterion) are evaluated, so **grid points are automatically added where oscillations or decay are rapid**.

References: src/grid/grid\_refine\_m.fypp:78-80, 214-226

## 2.4 Polytropic model treatment

The Lane–Emden solver uses a **fourth-order Taylor series + LSODAR event detection**:

```
X_BEG = sqrt(EPSILON(0._RD))      ! ~ 1e-8
y_exp(1) = 1 - x_exp**2/6 + n_poly*x_exp**4/120  ! theta(z) Taylor expansion
```

The key trick: the Lane–Emden RHS has a  $1/x^2$  factor that looks singular at  $x = 0$ , but using  $\theta \propto 1 - z^2/6$  near zero makes it finite.

Reference: src/poly/lane\_emden\_m.fypp:59, 109-110

## 2.5 Discretization: Magnus exponential integrator

No Chebyshev or FEM. GYRE uses **Gauss–Legendre Magnus integrators** (GL2 / GL4 / GL6):

```
! Diagonalize the ODE as dy_i/dx = lambda_i * y_i, then integrate exponentially
```

Reference: src/diff/magnus\_gl2\_diff\_m.fypp:20-56

## 2.6 Lessons from GYRE

| Technique                                                             | Portability to this project                    |
|-----------------------------------------------------------------------|------------------------------------------------|
| <b>Pre-designed non-dimensional variables</b> that absorb singularity | ★ ★ ★ Most central; worth adopting             |
| Analytic asymptotic BC (Dziembowski)                                  | ★★ Useful reference; requires physical insight |
| ODE-eigenvalue adaptive grid                                          | ★ Complex; not urgent                          |
| Magnus exponential integrator                                         | – Orthogonal to spectral methods               |

### 3. Dedalus v3's strategy

#### 3.1 Overall philosophy: weighted Jacobi bases

All of Dedalus's radial bases belong to the Jacobi family  $J_n^{(a,b)}(x)$ . The basis functions carry an intrinsic weight  $(1-x)^a(1+x)^b$ .

**Key insight.** Endpoint singularities can be **fully encoded in the weight exponents**  $a, b$ .

- ChebyshevT:  $a = b = -1/2$ , absorbs  $(1 \pm x)^{-1/2}$  singularities.
- Ultraspherical(alpha):  $a = b = \alpha - 1/2$ , tunable.
- Jacobi(a,b): arbitrary  $(a, b)$ .

*Reference:* dedalus/core/basis.py:435-515

#### 3.2 Ball/Disk basis: regularity at $r = 0$

BallBasis and DiskBasis use a **k parameter** to encode an  $r^k$  embedding:

```
# BallRadialBasis.__init__(..., k=2):  
#   basis functions automatically carry an  $r^k$  factor  
#   Zernike polynomials *  $r^{(\alpha + k + \ell + \text{regtotal})}$ 
```

regularity\_allowed() automatically filters out physically disallowed  $(\ell, m, \text{regularity})$  combinations.

*Reference:* dedalus/core/basis.py:3917-4087, 3531-3594

#### 3.3 Boundary conditions: tau / lift method

Instead of Dirichlet, Dedalus uses a lift method:

```
problem.add_equation("lap(f) + lift(tau) = -f**n")  
problem.add_equation("f(r=1) = 0") # the tau term absorbs boundary residual
```

LiftJacobi.build\_polynomial() picks a high-order polynomial  $\phi_N$ ; the boundary residual is projected onto it, avoiding pollution of the eigen-space.

*References:* dedalus/core/basis.py:790-815, examples/nlbvp\_ball\_lane\_emden/lane\_emden.py:67-71

#### 3.4 Variable-coefficient operator matrices

operator\_matrix() pre-computes variable-coefficient operators on Zernike / Jacobi bases, including  $r^{-k}$  singular operators, without manual regularization:

*Reference:* dedalus/core/basis.py:4044-4056

### 3.5 Things Dedalus does not do

- No Kosloff–Tal-Ezer (coordinate stretching) mapping.
- No Liouville  $\sqrt{\rho}$  substitution.
- No explicit cutoff.

Everything relies on the Jacobi weight absorbing singular behavior.

### 3.6 Lessons from Dedalus

| Technique                                                   | Portability to this project                                  |
|-------------------------------------------------------------|--------------------------------------------------------------|
| <b>Jacobi basis with chosen <math>(a, b)</math> weights</b> | ★ ★ ★ Core idea; structurally similar to GYRE’s substitution |
| $r^k$ embedding for Ball/Disk                               | ★★ Not immediately relevant (we’re 2D plane)                 |
| Tau / lift method for BC                                    | ★★ Cleaner than FD Dirichlet                                 |
| Variable-coefficient operator caching                       | ★ Engineering; later phase                                   |

## 4. Comparison: SL vs GYRE vs Dedalus

### 4.1 Three different philosophies for “variable coefficient + singular boundary”

| Aspect                 | Project SL (current)                   | GYRE                              | Dedalus                       |
|------------------------|----------------------------------------|-----------------------------------|-------------------------------|
| Discretization         | Interior FD                            | Magnus exponential                | Jacobi spectral               |
| Variable substitution  | $\hat{p} = \sqrt{\rho} q$              | $y_1 = x^{2-\ell} \xi / r$        | None (basis carries weight)   |
| Singularity absorption | <b>None; truncate</b><br>$\rho > 0.01$ | Variable definition absorbs       | $(1 - x)^a (1 + x)^b$ absorbs |
| Boundary condition     | Dirichlet (interior FD)                | Analytic asymptotic (Dziembowski) | Tau / lift                    |
| Surface singularity    | <b>Regexp to truncation</b>            | Variables finite at surface       | Basis handles intrinsically   |
| Convergence order      | Algebraic $\sim -2.4$                  | Analytic (Magnus high-order)      | Exponential                   |

### 4.2 Why is the Liouville $\sqrt{\rho}$ substitution not enough?

The Liouville substitution converts

$$\frac{d}{dy} \left[ \frac{1}{\rho} \frac{dp}{dy} \right] = f$$

into

$$q'' + Wq = g,$$

but  $W = \rho''/(2\rho) - 3(\rho')^2/(4\rho^2)$  **still diverges as  $\rho \rightarrow 0$** . We have only moved the singularity; we have not destroyed it.

GYRE and Dedalus instead **ensure no singular term ever appears** in the discretized operator:

- GYRE: the definitions  $y_1 = x^{2-\ell}\xi/r$  and  $y_2 = x^{2-\ell}P'/( \rho g r)$  force the entire LHS to be finite at the surface.
- Dedalus: Jacobi basis  $J_n^{(\alpha,\beta)}(x)$  carries  $(1-x)^\alpha$  factors, and if  $\alpha$  matches the decay exponent of  $\rho$ , the variable-coefficient ODE is analytically solvable in the weighted inner product.

## 5. Three possible paths forward for this project

### 5.1 Path A: keep Liouville SL, add a GYRE-style substitution

*Motivation.* Preserve the SL selling point: one set of  $(\mu_n, \psi_n)$  serving both Poisson inversion and the g-mode spectrum.

*How.* In addition to  $\hat{p} = \sqrt{\rho} q$ , add a second substitution  $\tilde{\phi} = r^\beta \phi$ , choosing  $\beta$  so that the induced operator  $\tilde{W}$  remains finite as  $\rho \rightarrow 0$ .

*Pros.* Physical elegance preserved; the g-mode spectrum still emerges automatically as an eigenspectrum of the basis.

*Cons.* Derivation-heavy; requires careful verification that  $\tilde{W}$  is smooth.

### 5.2 Path B: switch to Jacobi–Galerkin

*Motivation.* The Dedalus approach is numerically validated by our own Phase 0 E3 (smooth Gaussian  $\rho$  restores exponential convergence).

*How.* Drop the Liouville substitution and work directly in a Jacobi basis  $J_n^{(\alpha,\beta)}(y)$ . Pick  $\alpha = 1.5$  to match the Lane–Emden  $n = 1.5$  surface decay exponent.

*Pros.* Simplest engineering; Dedalus can be imported as a library for validation.

*Cons.* The g-mode spectrum now requires a **separate** generalized eigenproblem  $L\phi = \omega^2 M\phi$ . It is no longer a by-product of the basis. The “unified basis” pitch of the original design weakens.

### 5.3 Path C: do both A and B, write a method-comparison paper

*Motivation.* The numerical comparison is itself a contribution.

*How.*

1. Phase 1 in parallel: Boussinesq baseline using Liouville SL and Jacobi.
2. Compare  $\text{err}_{L^2}$ , wall-clock, and g-mode accuracy.
3. Publish a pure method-comparison paper (JCP fits well).

*Pros.* Most defensible; results cannot be wrong.

*Cons.*  $\sim 1.5$ – $2\times$  the workload; less focused narrative.

## 6. Recommendations

### 6.1 Path selection is a function of the paper’s angle

| Paper angle                                                             | Recommended path           |
|-------------------------------------------------------------------------|----------------------------|
| “GPU-efficient variable-coefficient Poisson” (JCP methods)              | Path B (Jacobi)            |
| “Unified spectral basis for Poisson + g-mode spectroscopy” (ApJS astro) | Path A (SL + substitution) |
| “Method comparison for stratified flows” (JCP comparison)               | Path C                     |

### 6.2 Technical risk assessment

**Path A risk.** (*Original 2026-05-02*) The substitution  $\phi \rightarrow r^\beta \phi$  requires  $\beta$  to match the Lane–Emden surface decay exponent of 1.5 with high accuracy. If mismatched, the singularity residue remains and the convergence is still algebraic. **Recommended: run a Python verification first to confirm that  $\beta = 1.5$  restores exponential convergence.**

**Updated 2026-05-03 (U1):** Path A  $\phi \rightarrow r^\beta \phi$   $\sigma \notin \mathbb{Z}$ .  $\beta = 1/4, -1/2, -3/4$  SymPy  $\alpha_2^* = 1 - \sigma/2, \sigma = 3/2$   $N^{-2}$ .  $(1-r)^{3/2}$  Chebyshev (Trefethen Thm 7.2,  $|a_n| \sim n^{-\sigma-1/2}$ ). Path A  $\sigma$ , raw Chebyshev.

**Path B risk.** Dedalus’s EVP solver may need careful tau-parameter tuning on stiff variable-coefficient problems. That said, Dedalus ships a Lane–Emden Ball example (`examples/nlbvp_ball_lane_emden/`), indicating the framework already handles analogous singularities. Engineering risk is low.

**Path C risk.** Largest workload, but risk is distributed.

### 6.3 Suggested execution order

#### Phase 0 ext+ (1–2 days, pure Python)

1. **Use Dedalus as a library, solve Lane–Emden Poisson** (quick validation of Path B).
  - Expected:  $\text{err}_{L^2} \leq 10^{-10}$ , exponential convergence.
  - If confirmed: Path B has zero technical risk.
2. **Keep Liouville SL, add  $\phi \rightarrow r^\beta \phi$  substitution** (Path A exploration).
  - Verify that matching  $\beta$  restores exponential convergence.
  - If confirmed: Path A has zero technical risk.
3. **Both results on the table, then decide Phase 1’s main path.**

This means Phase 1’s code commitment will be backed by numerical evidence, not by a hunch.



## 7. Conclusions

### 7.1 Main findings

1. **Neither GYRE nor Dedalus uses a Liouville  $\sqrt{\rho}$  substitution.** They absorb the singularity at the variable-definition or basis-weight level, from the start, rather than regularizing it after the fact.
2. **The Phase 0 algebraic —2.4 convergence is fixable.** E3 already demonstrated exponential convergence for a smooth Gaussian  $\rho$ . The problem is that our basis / variable substitution is not aligned with the singularity, not that the SL method itself is fundamentally limited.
3. **The g-mode selling point behaves very differently in each path.**
  - Liouville SL: g-mode spectrum is a free by-product (elegant).
  - Jacobi–Galerkin: g-mode requires a separate EVP (standard but loses the “unified basis” pitch).

### 7.2 Corrections to the original SL–anelastic design

The mathematics in docs/anelastic\_SL\_spectral\_design.md is **entirely correct** (the Liouville derivation in sect. 3–4 and the SL diagonalization in sect. 4.2 are both rigorous). The engineering assumptions, however, are **overly optimistic**:

- Section 8.2’s “smooth  $W \rightarrow$  exponential, singular  $W \rightarrow$  algebraic” dichotomy is now empirically confirmed by Phase 0.
- Section 8.1’s surface-singularity TODO has been Phase 0-confirmed as the primary convergence obstacle.
- Section 7’s “three-community gap” narrative remains correct, but the reason needs revision: it is not “no one thought of SL”; it is **“other communities have already adopted simpler approaches that sidestep the Liouville substitution”**.

Proposed revisions:

- Upgrade sections 8.1 and 8.2 from TODOs to **primary challenges**.
- Split Phase 1 into B-quick-validation and A-deep-exploration in parallel.
- Adjust the paper angle from “SL is the optimal method” to “SL vs Jacobi: a method comparison”.

### 7.3 Direction forward

**Updated 2026-05-03** (□□□ U3, U5). Phase 0 ext+ □□□, □□□ □□□□:

□□□ **(Phase 0 ext+, 2026-05-02..03)**: □□ raw Chebyshev collocation □ Lane-Emden  $n = 3$  □ GYRE full-gravity 4-var adiabatic □□ benchmark,  $N = 48$  (192 DOF) □□ max\_rd  $1.5 \times 10^{-6}$  vs GYRE,  $21 \times$  □□ staggered FD  $N_r = 1024$  □ DOF. Dedalus Path B □□□□(□□□□).

□□□ **(Phase 1, □□□)**: 2D Fourier-Chebyshev Boussinesq □□□. x □□ Fourier(□□ pseudo\_spectral cuFFT), y □□ raw Chebyshev collocation □ Eddington  $n = 3$  □□□. Rayleigh–Bénard Nu-Ra baseline.

□□ **(Phase 2)**: Boussinesq  $\rightarrow$  anelastic □□, □□□□□□  $\rho_0(y)$  □ continuity □□. □□  $n = 3/2$  □□□□□□ operationally □□, □□□□□□ Jacobi □□□□(Path B).

□□ **(Phase 3)**: □□□□□□ EVP □ g-mode/p-mode, □□□□□□□□ □□□□□□□□ runtime diagnostic — □□□□□□□□ **novelty** (□□-□□□□□□ 2D □□□□ DNS + □□□□□□□□).

(□□) □□□□: execute section 6.3 – Dedalus Lane–Emden validation plus Liouville +  $r^\beta$  substitution experiments.  $\rightarrow$  Dedalus □□□□□□(raw Chebyshev □  $n = 3$  □□□□);  $r^\beta$  □□□□□□(E6), □□□□□□□□  $\sigma$  □□.

## Appendix A: Key GYRE files

- `src/eqns/rad/rad_obound_m.fypp` – outer boundary condition dispatch
- `src/eqns/rad/rad_vars_m.fypp` – non-dimensional variables  $y_1, y_2$
- `src/grid/grid_refine_m.fypp` – adaptive grid refinement
- `src/poly/lane_emden_m.fypp` – Lane–Emden integrator
- `src/diff/magnus_gl2_diff_m.fypp` – Magnus GL2/GL4/GL6
- `docs/source/ref-guide/osc-equations/dimless-form.rst` – theoretical formulation

## Appendix B: Key Dedalus v3 files

- `dedalus/core/basis.py:435-515` – Jacobi / ChebyshevT / Ultraspherical
- `dedalus/core/basis.py:3917-4087` – BallBasis / DiskBasis
- `dedalus/core/basis.py:790-815` – Lift / Tau method
- `dedalus/core/basis.py:4044-4056` – variable-coefficient operator matrices
- `examples/nlbvp_ball_lane_emden/lane_emden.py` – Lane–Emden example
- `examples/evp_disk_pipe_flow/pipe_flow.py` – Disk EVP example

## Appendix C: References

- **Dziembowski (1971)**: Non-radial oscillations of evolved stars. I. *Acta Astronomica* 21, 289.
- **Unno et al. (1989)**: *Nonradial Oscillations of Stars*, 2nd ed., University of Tokyo Press.
- **Tassoul (1980)**: Asymptotic approximations for stellar nonradial pulsations. *ApJS* 43, 469.
- **Townsend & Teitler (2013)**: GYRE: an open-source stellar oscillation code. *MNRAS* 435, 3406.
- **Burns et al. (2020)**: Dedalus: a flexible framework for numerical simulations with spectral methods. *PhRvR* 2, 023068.
- **Kosloff & Tal-Ezer (1993)**: A modified Chebyshev pseudospectral method. *JCP* 104, 457.
- **Boyd (2001)**: *Chebyshev and Fourier Spectral Methods*. Dover Publications.